

MARABA, Brazil

WMO: 825620

Lat: 5.3661S

Lon: 49.1252W

Elev: 334

StdP: 14.52

Time Zone: -3.00 (W03)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	68.1	69.8	60.6	79.9	92.4	62.6	85.9	88.4	12.4	79.3	10.7	79.5	1.5	200	0.317

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	22.2	100.6	77.4	98.8	77.1	97.1	77.4	83.2	91.2	82.1	90.6	81.4	89.9	6.3	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.0	162.8	86.7	80.4	159.5	86.0	79.1	153.1	84.4	47.3	91.9	46.3	91.2	45.5	89.9	95.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.3	9.7	8.7	DB	65.2	102.6	1.8	2.9	63.9	104.7	62.8	106.4	61.8	108.0	60.4	110.2	(4)
(5)			WB	63.3	86.1	1.5	4.0	62.3	89.0	61.4	91.3	60.6	93.6	59.5	96.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.5	81.9	81.7	82.5	83.0	83.3	83.3	83.3	84.7	86.0	85.4	84.0	82.8
	DBStd	2.51	2.47	2.35	2.17	2.16	1.84	1.70	1.79	1.98	2.19	2.63	2.32	2.27
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12229	990	889	1006	990	1034	1000	1033	1075	1079	1098	1020	1015
	CDD65	6754	525	469	541	540	569	550	568	610	629	633	570	550
	CDH74	69824	4711	4088	4909	5100	5635	5757	6191	7119	7477	7338	6173	5326
	CDH80	33008	1796	1525	1851	2000	2495	3069	3545	4070	4014	3672	2791	2180
Wind	WSAvg	3.4	3.3	3.3	3.4	3.1	2.8	2.9	3.1	3.9	4.5	4.2	3.5	3.2
Precipitation	PrecAvg	72.6	11.8	12.5	14.6	10.1	3.8	1.1	0.7	0.5	1.7	3.5	5.6	9.1
	PrecMax	94.1	22.5	24.2	24.3	18.2	8.3	3.1	4.2	1.8	4.0	8.6	13.5	20.5
	PrecMin	54.7	2.3	3.9	4.1	1.4	0.2	0.0	0.0	0.0	0.2	0.2	1.8	5.0
	PrecStd	12.3	5.0	5.6	4.3	4.5	2.2	0.9	0.9	0.5	1.1	2.2	2.8	3.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	95.2	95.2	95.4	96.9	97.1	99.0	101.8	102.5	103.8	102.3	98.2	96.7
		MCWB	80.6	80.5	81.3	81.4	81.3	78.0	74.8	76.8	76.2	78.8	79.9	81.6
	2%	DB	93.0	93.0	93.1	93.6	94.9	97.0	98.7	100.2	100.4	98.6	95.4	93.4
		MCWB	80.5	80.5	81.0	81.1	80.3	78.3	75.9	76.1	77.7	79.3	79.8	80.0
	5%	DB	91.1	90.8	91.2	91.7	93.1	95.0	96.7	98.3	98.3	96.5	93.5	91.6
		MCWB	80.0	80.1	80.5	80.4	80.2	77.7	75.8	75.7	77.3	78.7	79.5	79.6
	10%	DB	89.2	88.1	89.2	89.7	91.3	93.2	94.9	96.6	96.4	94.6	91.7	89.9
		MCWB	79.6	79.6	79.9	79.9	79.5	77.9	76.0	75.6	77.3	78.7	79.2	79.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.3	83.5	83.5	83.8	84.3	83.8	80.8	80.8	81.8	82.5	82.8	83.7
		MCDB	89.4	89.6	91.4	91.5	92.2	94.2	91.6	94.7	93.4	91.7	91.9	90.1
	2%	WB	81.5	81.5	81.9	82.2	81.9	81.0	79.6	79.6	80.0	80.8	81.4	81.5
		MCDB	89.0	89.2	89.6	90.3	90.5	90.9	91.2	93.4	92.9	92.2	91.0	89.5
	5%	WB	80.6	80.5	81.0	81.3	80.9	79.9	78.7	78.7	79.2	80.0	80.6	80.5
		MCDB	88.0	87.9	88.4	88.9	89.3	89.8	90.5	92.2	92.3	91.4	89.8	88.6
	10%	WB	79.7	79.6	80.1	80.4	80.1	79.1	77.8	77.8	78.5	79.2	79.8	79.8
		MCDB	86.4	86.1	86.7	87.5	87.9	88.9	89.3	90.5	91.4	90.2	88.4	87.1
Mean Daily Temperature Range	5% DB	MDBR	15.2	15.0	15.3	15.8	17.4	22.0	25.1	25.6	22.2	19.2	17.0	15.9
		MCDBR	18.0	18.0	17.8	18.5	20.1	24.9	28.0	28.5	25.4	22.2	19.4	18.2
		MCWBR	7.1	7.3	6.9	7.0	7.3	8.2	8.5	8.5	7.1	6.0	6.1	6.6
	5% WB	MCDBR	16.2	16.4	16.2	16.9	17.9	20.7	23.3	24.2	22.2	19.5	17.7	16.9
		MCWBR	7.2	7.3	6.9	7.1	7.2	7.8	8.1	7.9	6.8	6.4	6.3	6.9
Clear-Sky Solar Irradiance	taub	0.400	0.394	0.391	0.392	0.402	0.381	0.390	0.423	0.519	0.553	0.528	0.466	
	taud	2.491	2.521	2.536	2.529	2.480	2.527	2.472	2.340	2.074	2.004	2.081	2.282	
	Ebn at Noon	297	298	295	287	276	277	276	274	256	251	259	276	
	Edh at Noon	36	36	35	34	34	31	34	40	54	59	55	45	
All-Sky Solar Radiation	RadAvg	1463	1492	1524	1530	1587	1686	1771	1834	1681	1548	1473	1445	
	RadStd	112	111	51	65	84	69	80	69	102	89	70	67	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page